

Artificial and Computational Intelligence

Assignment – I

Name of Student	BITS ID
Mallidi Akhil Reddy	2024TM93056
Ashwarya Anupam	2024TM93057
Vaidya Rucha Sandeep	2024TM93058
Shivam Prabhakar	2024TM93059
J Rakesh	2024TM93061

GPS Navigation Agent

❖ Explain the PEAS (Performance measure, Environment, Actuator, Sensor) for the Agent

1. Performance (P)

The agent's goal is to guide the user from Office (start state) to Home (goal state) by finding the safest and shortest path.

Performance is measured by:

- Reaching the goal (Home) successfully without getting stuck.
- Avoiding obstacles (road blocks, buildings)
- Maximizing safety score
- Minimizing number of steps (shortest path length)
- Ensure successful completion without getting stuck.

2. Environment (E)

The environment is the city grid world.

- Intersections (grid nodes)
- Road Paths
- Blockages or road closure
- Traffic
- Buildings

Characteristics of the Environment

- Deterministic (same action → same result)
- Static (no dynamic changes in grid while navigating)
- Fully observable (the GPS knows the whole map)

3. Actuators (A)

Movement commands to navigate the grid:

- Move Up
- Move Down
- Move Right
- Move Left

4. Sensors (S)

- Detect agent's current position – GPS receiver
- Sense the adjacent cells if they are free, blocked or contain a building – Camera / Radar / Lidar
- Monitor path length – Odometer

- Measure distance between intersections and known static hazards - Digital Map

	0	1	2	3	4	5
0	 Office					
1						
2						
3						
4						
5						 Home

Co-ordinates

- Office → (0,0)
- Home → (5,5)
- School → (1,0)
- Classroom → (3,0)
- Meeting → (5,1)
- Warehouse → (3,5)
- Road Blocks → [(0,2), (0,3)] ; (1,2) ; (2,4) ; (3,2) ; (5,3)

Path Cost

- +1 per move
- +5 move adjacent to building
- -3 move adjacent to roadblock

No Diagonal Movement Allowed

But for calculating cost, we have considered the buildings and roadblocks which are diagonal to the cell.

❖ Recursive Best First Search

$$f(n) = g(n) + h(n)$$

$g(n) \rightarrow$ Cost Function

$h(n) \rightarrow$ Heuristic Fun = Manhattan distance

1. Initial State $\rightarrow$ Office – (0,0)

Moves possible $\rightarrow$ (1,0), (0,1)

$$g(0,0) = 0$$

$$h(0,0) = 10 \quad [= |5-0| + |5-0|]$$

$$\mathbf{f(0,0) = 0 + 10 = 10}$$

Expand neighbor (0,1):

(Note: (1,0) is School, which is a building — we cannot expand into it)

So we skip (1,0) entirely — no g , h , f , no expansion.

Thus from (0,0), the only valid expansion is into (0,1).

Current state -- (0,1)

Moves possible $\rightarrow$ (1,1)

Compute cost:

$$g(0,1) = 1 + 5 + 5 - 3 - 3 = 5$$

$$h(0,1) = |5-1| + |5-0| = 4 + 5 = 9$$

$$f(0,1) = 5 + 9 = 14$$

2. Current state -- (0,1)

Moves possible $\rightarrow$ (1,1)

Compute cost:

$$g(1,1) = 1 + 5 + 5 - 3 - 3 = 5$$

$$h(1,1) = |5-1| + |5-1| = 4 + 4 = 8$$

$$f(1,1) = 5 + 8 = 13$$

3. Current state -- (1,1)

Moves possible $\rightarrow$ (2,1)

Compute cost:

$$g(2,1) = 1 + 5 + 5 - 3 - 3 = 5$$

$$h(2,1) = |5-2| + |5-1| = 3 + 4 = 7$$

$$f(2,1) = 5 + 7 = 12$$

4. Current State – (2,1)

Moves Possible – (2,0), (2,2), (3,1)

$$g(2,0) = 1 + 5 + 5 = 11$$

$$h(2,0) = |5-0| + |5-2| = 5 + 3 = 8$$

$$f(2,0) = 11 + 8 = 19$$

$$g(2,2) = 1 + -3 - 3 = -5$$

$$h(2,2) = 3 + 3 = 6$$

$$\mathbf{f(2,2) = -5 + 6 = 1}$$

$$g(3,1) = 1+5-3=3$$

$$h(3,1) = 2+4=6$$

$$f(3,1) = 3+6=9$$

thus selecting (2,2) as the best way ahead.

5. Current State – (2,2)

Moves possible – (2,3)

At (2,3) surrounded by 3 road block present at (1,2) , (3,2) , (2,4).

$$g(2,3) = 1 - 3 - 3 - 3 = -8$$

$$h(2,3) = 2 + 3 = 5$$

$$f(2,3) = -8 + 5 = -3$$

6. Current State – (3,2)

Moves Possible – A (3,3) & B (1,3)

$$g(3,3) = 1 - 3 - 3 = -5$$

$$h(3,3) = 2 + 2 = 4$$

$$f(3,3) = -5 + 4 = -1$$

$$g(1,3) = 1 - 3 - 3 - 3 = -8$$

$$h(1,3) = 4 + 2 = 6$$

$$f(1,3) = -8 + 6 = -2$$

Let's Consider B(1,3) as $f(1,3) = -2$, Next Moves possible from B is (1,4)

$$g(1,4) = 1 - 3 - 3 = -5$$

$$h(1,4) = 4 + 1 = 5$$

$$f(1,4) = -5 + 5 = 0$$

Let's Consider A(3,3) as $f(3,3) = -1$, Next Moves possible from A is (3,4) & (4,3)

$$g(3,4) = 1 + 5 - 3 = 3$$

$$h(3,4) = 1 + 2 = 3$$

$$f(3,4) = 3 + 3 = 6$$

$$g(4,3) = 1 - 3 - 3 = -5$$

$$h(4,3) = 2 + 1 = 3$$

$$\mathbf{f(4,3) = -5 + 3 = -2}$$

We take route A through (3,3) to reach (4,3)

7. Current State – (4,3)

Moves possible – A (4,4) & B (4,2)

$$g(4,2) = 1+5-3-3 = 0$$

$$h(4,2) = 1+3=4$$

$$f(4,2) = 0+4=4$$

$$g(4,4) = 1+5-3=3$$

$$h(4,4) = 1+1=2$$

$$f(4,4) = 3+2=5$$

Let's Consider B(4,2) as $f(4,2) = 4$, Next Moves possible from B is (4,1) & (5,2)

$$g(4,1) = 1 + 5 + 5 - 3 = 8$$

$$h(4,1) = 1+4=5$$

$$f(4,1) = 8+5=13$$

$$g(5,2) = 1 + 5 - 3 = 3$$

$$h(5,2) = 0 + 3=3$$

$$f(5,2) = 3+3= 6$$

Next Let's Consider A (4,4) as $f(4,4) = 5$, Next Moves possible from A is (4,5) & (5,4)

$$g(5,4) = 1 + 5 - 3 = 3$$

$$h(5,4) = 0 + 1 = 1$$

$$\mathbf{f(5,4) = 3 + 1 = 4}$$

$$g(4,5) = 1 + 5 + 5 = 11$$

$$h(4,5) = 1+0=1$$

$$f(4,5) = 11+1=12$$

We take route A through (4,4) to reach (5,4)

8. Current State – (5,4)

Moves possible – (5,5) → Goal

$$g(5,5) = 1$$

$$h(5,5) = 0$$

$$f(5,5) = 1+0=1$$

Thus optimal path is

$(0,0) \rightarrow (0,1) \rightarrow (1,1) \rightarrow (2,1) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (3,3) \rightarrow (4,3) \rightarrow (4,4) \rightarrow (5,4) \rightarrow (5,5)$

❖ Uniform Cost Search

Path or nodes navigate to reach the goals

$$1. (0,1) = 1+5+5-3-3 \\ = 5$$

$$2. (1,1) = 1+5+5-3-3 \\ = 5$$

$$3. (2,1) = 1+5+5-3-3 \\ = 5$$

$$4. (2,2) = 1-3-3-5 \\ = -5$$

$$5. (2,3) = 1-3-3-3$$

$$= -8$$

$$(1,3) = 1-3-3-3$$

$$= -8$$

Here we treated roadblocks at (0,2) (0,3) as 1 road block

$$6. (1,4) = 1-3-3$$

$$= -5$$

Now we have one problem here, (0,4) and (1,5) = -2

As our cost is higher than previously visited node (1,4) = -5

$$7. (1,4) = -5$$

$$8. (3,3) = 1-3-3$$

$$= -5$$

$$9. (4,3) = 1-3-3$$

$$= -5$$

$$10. (4,2) = 1-3-3+5$$

$$= 0$$

$$11. (4,4) = 1-3+5+5$$

$$= 8$$

$$12. (5,4) = 1-3+5$$

$$= 3$$

$$13. (5,5) = 1$$

Thus goal is reached.